

Far Eastern University

# PREDICTING THE NEXT WAVE: A SHALLOW WATER EQUATION-BASED PHYSICS-INFORMED NEURAL NETWORK


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**Bachelor of Science in Applied Mathematics**

**(Information Technology Track)**

May 2022

# Declaration of Authorship

We, Elisha Sophia S. Borromeo, Jan Edward F. Macagba, and Consuelo B. Mercado, declare that this thesis titled, Predicting the Next Wave: A Shallow Water Equation-Based Physics-Informed Neural Network , and the work present in it are our own. We confirm that:

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- Where we have consulted the published work of others, this is always clearly attributed.
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# Abstract

## Predicting the Next Wave: A Shallow Water Equation-Based Physics-Informed Neural Network

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Flooding is a recurring problem in highly urbanized areas of Metro Manila, including Pasig City, where heavy rainfall, limited drainage capacity, and changes in land use contribute to the accumulation of surface water. Traditional hydrodynamic models solve the governing equations numerically but may demand substantial computational resources and detailed environmental data, while purely data-driven machine learning models may not satisfy the physical laws governing water flow. This study develops and evaluates a Shallow Water Equation-based Physics-Informed Neural Network (SWE-based PINN) for flood modeling in a selected area of Pasig City.

The Shallow Water Equations, which express the conservation of mass and momentum in depth-averaged flow, serve as the governing mathematical model. They are incorporated into the training of the neural network through a composite loss function that combines data, governing-equation residual, initial condition, and boundary condition terms, with the residuals computed by automatic differentiation. The network takes spatial, temporal, rainfall, terrain, and simplified drainage variables as inputs and predicts water depth and horizontal water velocity. The study follows a quantitative, computational, and model-development design that uses secondary data such as rainfall records,

water-level and discharge measurements, digital elevation models, and historical flood observations.

The performance of the model is assessed in terms of convergence, approximation error, and physical consistency, particularly the satisfaction of mass and momentum conservation. Evaluation metrics include root mean square error, mean absolute error, the coefficient of determination, and physics-based residual error. The predictions of the PINN are compared with a classical numerical method for solving the Shallow Water Equations. The findings are limited to the selected study area and the available datasets, and the model is not intended to replace official flood forecasting systems. The study aims to examine the applicability of a physics-informed computational approach to urban flood modeling and to serve as a foundation for future research on PINNs and the Shallow Water Equations.

# Acknowledgments

Type your acknowledgements here. It is customary to acknowledge special assistance from extramural agencies. There is no obligation that assistance received from members of the dissertation or thesis committee be acknowledged.
Acknowledgments should be couched in terms consistent with the scholarly nature of the work. Your name and the date should not appear on this page.

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# Chapter 1

# The Problem and Its Background

## 1.1 Background of the Study

Flooding remains one of the major environmental challenges in the Philippines, particularly in highly urbanized areas where heavy rainfall, limited drainage capacity, and changes in land use contribute to the accumulation of surface water. In Metro Manila, flooding continues to affect residential communities, transportation systems, commercial establishments, and public infrastructure. Pasig City, which is situated near major waterways such as the Pasig River and is surrounded by densely developed urban areas, is among the locations that may experience flooding during periods of intense rainfall. These conditions highlight the importance of mathematical and computational approaches that can help predict flood behavior and improve the understanding of water movement in urban environments.

Flood prediction plays an important role in disaster risk reduction and urban planning. Accurate estimates of water depth and flow velocity can provide valuable information for identifying flood-prone areas and understanding the possible progression of inundation. Traditional hydrodynamic models simulate water movement by numerically solving governing mathematical equations. Although these models can be accurate, they are computationally slow and data-intensive (Liu et al., 2026). Guo, Guan, and Yu (2021) noted that computational efficiency, model complexity, and the availability of high-quality input data are important challenges in urban flood modeling. The practical consequence of this cost was illustrated by Besseling, Bomers, and Hulscher (2024), who reported that simulating flood spread over a large area with a hydrodynamic model can take 6 to 10 hours, which is too long when authorities need predictions within seconds

or minutes to support evacuation decisions.

Data-driven machine learning models offer a faster alternative. For example, the Long Short-Term Memory (LSTM) network of Besseling et al. (2024) produced flood-depth predictions in a fraction of a second after training. However, such models depend heavily on their training data, may behave as "black boxes," and may produce predictions that violate physical principles (Liu et al., 2026; Perera, Angeloudis, Siripatana, et al., 2026). In their systematic review of hybrid flood inundation models, Perera et al. (2026) described this trade-off between the realism of physics-based models and the speed of machine learning models, and identified hybrid approaches as a way to combine the strengths of both.

The Shallow Water Equations (SWE) provide a mathematical representation of water movement by describing the conservation of mass and momentum in depth-averaged flow. These equations are widely used in flood modeling because they account for changes in water depth and horizontal velocity over space and time (Guo et al., 2021). However, solving the SWE numerically may involve computational challenges, especially when dealing with complex terrain and spatially varying conditions.

Physics-Informed Neural Networks (PINNs) are one such hybrid approach. Introduced by Raissi, Perdikaris, and Karniadakis (2019), PINNs incorporate governing differential equations into the loss function of a neural network, so that the network is trained to be consistent with both the available data and the governing physical laws. Perera et al. (2026) identified PINNs as especially promising because they embed physical laws directly into machine learning. PINNs have since been applied to flood problems. Liu et al. (2026) developed a Shallow Water Equation-based PINN for urban flood depth prediction in Hongshan District, Wuhan, China, and reported better performance than DNN, CNN, and LSTM models. Tian, Ding, Su, Huang, and Chen (2025) investigated PINNs for solving the two-dimensional SWE with terrain topography and rainfall source terms using benchmark cases. Praveen et al. (2024) developed a PINN for real-time flood prediction from rainfall-runoff data.

Despite these developments, important gaps remain. The framework of Liu et al. (2026) was developed for a single urban area and relied on simulated data, while the cases of Tian et al. (2025) were controlled benchmark problems rather than a specific real-world location. Praveen et al. (2024) used a publicly available dataset, so its real-world performance requires further validation. Perera et al. (2026) also noted that hybrid flood models are evaluated using different datasets, conditions, and metrics, which makes them difficult to compare, and proposed standardized benchmarking against traditional hydrodynamic models. Consequently, the convergence, approximation error, and physical consistency of a SWE-based PINN in Pasig City have not been established.

In this study, a Shallow Water Equation-based Physics-Informed Neural Network will be developed and implemented in a selected area of Pasig City. The proposed model will be evaluated in terms of its convergence, approximation error, and ability to satisfy physical constraints. Its performance will also be compared with a classical numerical method to determine its effectiveness in solving the Shallow Water Equations for flood modeling.

## 1.2 Statement of the Problem

Flooding remains a significant concern in urban areas such as Pasig City, where rainfall, terrain elevation, and water flow conditions influence flood behavior. Predicting the movement and depth of floodwater requires a mathematical model that can represent the physical processes governing water flow. The Shallow Water Equations (SWE) provide such a representation, but solving them with classical numerical methods can be computationally demanding and dependent on detailed input data, particularly in complex urban settings (Guo et al., 2021; Liu et al., 2026). Meanwhile, purely data-driven approaches can be fast but depend heavily on their training data and may not satisfy the physical laws governing water movement (Liu et al., 2026; Perera et al., 2026).

Physics-Informed Neural Networks (PINNs) combine neural network approximation

with governing physical equations (Raissi et al., 2019). Previous studies have applied PINNs to urban flood depth prediction (Liu et al., 2026), to the two-dimensional SWE with terrain topography and rainfall source terms (Tian et al., 2025), and to real-time flood prediction from rainfall-runoff data (Praveen et al., 2024). However, these studies were conducted using their own study areas, datasets, and model configurations. They relied on simulated, benchmark, or publicly available data, and none of them applied a SWE-based PINN to Pasig City. Systematic comparisons of PINN convergence and approximation error against classical numerical methods across conditions also remain limited (Tian et al., 2025; Perera et al., 2026). It therefore remains unestablished how a SWE-based PINN performs in Pasig City in terms of its convergence, approximation error, and ability to satisfy the conservation of mass and momentum.

The study will seek to answer the following general research question:

> How can a Shallow Water Equation-based Physics-Informed Neural Network be developed and evaluated for predicting flood behavior in Pasig City?

Specifically, the study will seek to answer the following questions:

1. How can the Shallow Water Equations be formulated as the governing mathematical model for flood and water flow behavior in Pasig City?
2. How can a Physics-Informed Neural Network be developed to incorporate the Shallow Water Equations into its training process?
3. What are the convergence behavior and approximation error of the proposed PINN in solving the Shallow Water Equations?
4. How does the performance of the PINN compare with that of a classical numerical method when evaluated using appropriate error metrics?
5. To what extent does the proposed PINN satisfy the physical constraints of water flow, particularly the conservation of water mass and momentum?

## 1.3 General and Specific Objectives

### 1.3.1 General Objective

The general objective of this study is to develop and evaluate a Shallow Water Equation-based Physics-Informed Neural Network for predicting flood behavior in Pasig City.

### 1.3.2 Specific Objectives

Specifically, this study aims to:

1. Formulate the Shallow Water Equations as the governing mathematical model for flood and water flow behavior in Pasig City.
2. Develop a Physics-Informed Neural Network that incorporates the Shallow Water Equations into its training process.
3. Evaluate the convergence and approximation error of the proposed PINN in solving the Shallow Water Equations.
4. Compare the performance of the PINN with a classical numerical method using appropriate error metrics.
5. Assess the model's ability to satisfy physical constraints, particularly the conservation of water mass and momentum.

## 1.4 Conceptual Framework

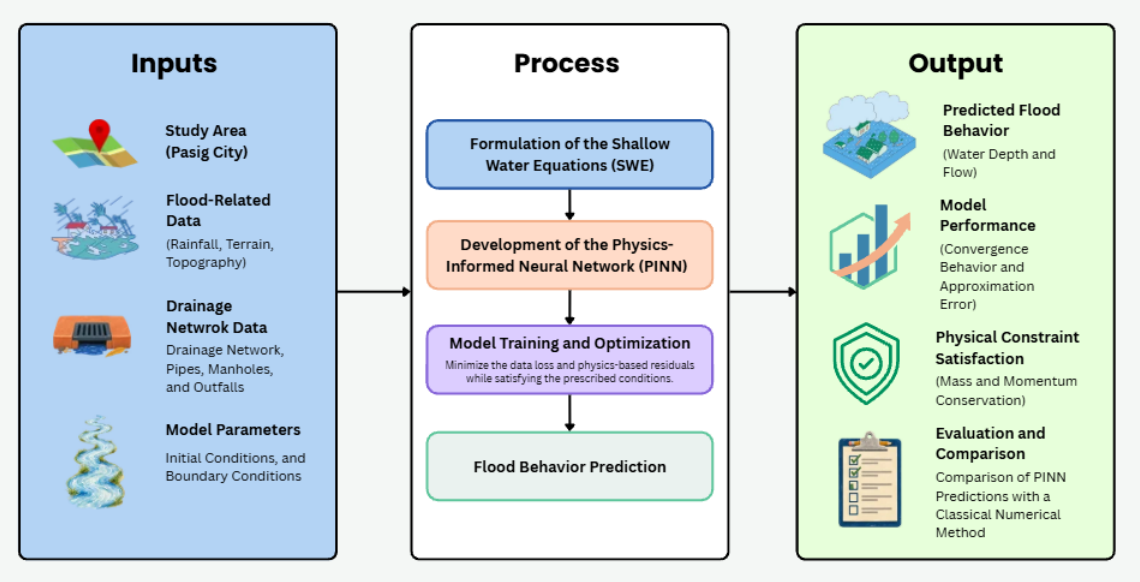



**Figure 1.1**: Conceptual Framework of the SWE-Based Physics-Informed Neural Network for Flood Modeling in Pasig City

The conceptual framework of the study follows an Input-Process-Output (IPO) model. It illustrates how the input data are processed through a Physics-Informed Neural Network incorporating the Shallow Water Equations to generate predictions of water depth, water flow, and flood behavior.

1. **Input Data:** The input data consist of rainfall data, terrain elevation, drainage network data, initial conditions, and boundary conditions for the selected study area in Pasig City. The drainage network data will provide information needed to incorporate the drainage rate into the governing equations. These inputs provide the environmental, spatial, and mathematical information required for the flood modeling process.

2. **Process:** The input data will be processed using a SWE-based Physics-Informed Neural Network (PINN). The governing Shallow Water Equations, including the

drainage rate as a representation of water removal through the drainage system, will be incorporated into the model's loss function as physical constraints. The network will undergo training and optimization by minimizing the data loss and the residuals of the governing equations while satisfying the specified initial and boundary conditions. Once trained, the model will be used to predict flood behavior across the selected study area.

3. **Output:** The proposed model will generate predictions of water depth and horizontal water velocity, representing the flood behavior across the selected study area in Pasig City. The model will also be evaluated based on its convergence behavior, approximation error, and ability to satisfy the physical constraints represented by the conservation of mass and momentum.

## 1.5 Significance of the Study

This study aims to contribute to the development and evaluation of physics-informed computational approaches for flood modeling. By integrating the Shallow Water Equations into a neural network, the research will examine how physical constraints can be incorporated into flood prediction and how the resulting model performs compared with a classical numerical method.

### 1.5.1 Local Government Units

The findings of the study may provide useful information to local government units, particularly Pasig City, regarding the potential application of physics-informed neural networks in flood modeling. The predicted water depth and velocity may help improve the understanding of flood behavior in selected areas. However, the model is intended as a research-based computational approach and will not replace official flood forecasting systems.

### 1.5.2 Disaster Risk Reduction and Management Offices

The study may contribute to the development of computational approaches that can support flood assessment and disaster risk reduction. The results may provide additional insights into the movement and accumulation of water during flooding, which could be useful in understanding flood-prone areas.

### 1.5.3 Researchers and the Academic Community

This study may serve as a reference for researchers interested in mathematical modeling, numerical solutions of partial differential equations, and Physics-Informed Neural Networks. The comparison between the PINN and a classical numerical method may contribute to further investigations into the accuracy, convergence, and physical consistency of physics-informed models.

### 1.5.4 Future Researchers and Students

The study may serve as a foundation for future research involving PINNs and the Shallow Water Equations. Future researchers may extend the model by incorporating more detailed terrain data, additional hydrological variables, or more complex representations of urban drainage systems.

## 1.6 Scope and Limitations

This study will focus on the development and evaluation of a Shallow Water Equation-based Physics-Informed Neural Network for flood modeling in Pasig City, Philippines. The research will utilize the Shallow Water Equations as the governing mathematical model to represent the conservation of mass and momentum in water flow. The PINN will be developed to predict water depth and horizontal water velocity based on the available rainfall data, terrain elevation, initial conditions, and boundary conditions.

The study will evaluate the performance of the proposed PINN by comparing its numerical predictions with those obtained from a classical numerical method. The comparison will focus on convergence, approximation error, and physical consistency. Appropriate error metrics will be used to assess the differences between the two methods.

The scope of the study is limited to the selected study area within Pasig City and the available datasets. The accuracy of the model may be affected by the quality, completeness, and spatial and temporal resolution of the rainfall and terrain data. The availability of observed flood depth and water velocity measurements may also limit the extent of model validation.

Furthermore, the mathematical model will rely on the assumptions associated with the Shallow Water Equations, including the representation of water flow as a depth-averaged process. The study will not fully represent every physical factor that may influence urban flooding, such as highly detailed drainage network behavior, localized obstructions, or complex interactions between surface water and underground infrastructure. The model's computational performance may also depend on the available hardware and training configuration.

The study will primarily assess the PINN as a mathematical and computational approach to solving the Shallow Water Equations. Therefore, the findings will not necessarily establish the model's readiness for real-time operational flood forecasting or its applicability to all flood-prone areas in the Philippines.

## 1.7 Definition of Terms

The following terms are defined conceptually and operationally to provide a clearer understanding of the concepts used in this study.

- **Shallow Water Equations (SWE).** The Shallow Water Equations are a system of partial differential equations derived from the principles of conservation of mass and momentum. In this study, they serve as the governing

mathematical model for representing water flow and flood behavior in Pasig City.

- **Physics-Informed Neural Network (PINN).** A Physics-Informed Neural Network is a neural network that incorporates governing physical equations into its training process through a loss function. In this study, the PINN will use the Shallow Water Equations as physical constraints to predict water depth and horizontal water velocity.
- **Flood Modeling.** Flood modeling refers to the mathematical and computational representation of flood behavior. In this study, it involves simulating water movement and estimating water depth and velocity within the selected study area.
- **Water Depth.** Water depth refers to the vertical distance between the ground surface and the water surface. In this study, it is one of the primary variables predicted by the SWE-based PINN.
- **Horizontal Water Velocity.** Horizontal water velocity refers to the rate and direction of water movement along the horizontal plane. In this study, it is represented by the velocity components in the $x$- and $y$-directions.
- **Initial Conditions.** Initial conditions specify the state of the modeled water system at the beginning of the simulation. In this study, they define the initial water depth and velocity values used in solving the Shallow Water Equations.
- **Boundary Conditions.** Boundary conditions describe the behavior of water flow along the boundaries of the computational domain. In this study, they define the constraints applied at the edges of the selected flood modeling area.
- **Approximation Error.** Approximation error refers to the difference between the solution produced by a computational method and a reference solution. In this study, it will be used to evaluate the accuracy of the PINN in comparison with the classical numerical method.

- **Convergence.** Convergence refers to the tendency of a numerical or computational solution to approach a stable or reference solution as the relevant approximation or training process progresses. In this study, it will be evaluated to examine the behavior of the PINN and the classical numerical method in solving the Shallow Water Equations.
- **Conservation of Mass.** Conservation of mass is the physical principle stating that mass is neither created nor destroyed within a system, although it may enter or leave through the system boundaries. In this study, it is represented by the continuity equation of the Shallow Water Equations.
- **Conservation of Momentum.** Conservation of momentum is the physical principle describing how the momentum of water changes due to forces such as gravity, friction, and flow interactions. In this study, it is represented by the momentum equations of the Shallow Water Equations.

# Chapter 2

# Review of Related Literature and Studies

This chapter presents the concepts, theories, literature, and previous studies relevant to the development of a Shallow Water Equation-based Physics-Informed Neural Network (PINN) for flood prediction in Pasig City. The discussion establishes the mathematical and computational foundation of the proposed model by examining flooding, hydrodynamic modeling, artificial neural networks, Physics-Informed Neural Networks, and the Shallow Water Equations. It also reviews previous applications of these approaches in flood modeling and identifies the research gap addressed by the present study.

## 2.1 Preliminaries

This section introduces the fundamental concepts necessary for understanding the proposed study. It discusses the characteristics of flooding, hydrodynamic modeling, artificial neural networks, and Physics-Informed Neural Networks. These concepts provide the foundation for understanding how the governing equations of water flow can be incorporated into a neural network for flood prediction.

### 2.1.1 Fundamental Concepts of Flooding

Flooding occurs when water accumulates in an area beyond its normal capacity to contain or drain it. It may result from heavy rainfall, overflowing rivers, inadequate drainage capacity, and changes in land use. In urban areas, the presence of buildings, roads, and other impermeable surfaces can reduce the infiltration of water into the ground, resulting in increased surface runoff. When the amount of runoff exceeds the capacity of the surrounding drainage system, water may accumulate on streets and other low-lying

areas.

Flood behavior is influenced by several hydrological and physical variables. Rainfall refers to the amount of water supplied to the ground through precipitation. Surface runoff is the portion of rainfall that flows over the ground instead of infiltrating into the soil. Terrain elevation influences the direction and distribution of water flow, while drainage systems provide pathways for collecting and removing excess water.

Water depth and flow velocity are important variables in flood modeling. Water depth refers to the vertical distance between the ground surface and the water surface. Flow velocity describes the speed and direction of water movement. In a two-dimensional flood model, velocity is commonly represented by two components, namely the velocity in the $x$-direction and the velocity in the $y$-direction.

These variables are important in understanding the development and movement of floodwater. In the present study, water depth and horizontal water velocity will be the primary variables predicted by the proposed SWE-based PINN.

### 2.1.2 Hydrodynamic Modeling

Hydrodynamic modeling refers to the mathematical and computational representation of water movement. It uses governing equations derived from physical principles to describe how water changes in depth, velocity, and distribution over space and time. Hydrodynamic models are commonly used in flood simulation, river flow analysis, coastal processes, and water resource management.

The movement of water can be described using mathematical equations such as the Navier-Stokes equations. However, these equations may be simplified under certain assumptions to represent shallow water flow. One commonly used formulation is the Shallow Water Equations, which describe the conservation of mass and momentum in depth-averaged water flow.

Traditional hydrodynamic models solve governing equations using numerical methods. Common approaches include finite difference, finite volume, and finite element methods.

These methods approximate the mathematical solution over a computational domain divided into discrete spatial and temporal points.

Although numerical methods are widely used in flood modeling, their computational requirements may increase when the study area is large or when detailed terrain and flow conditions are considered. Urban flood modeling may also require detailed information regarding terrain elevation, rainfall, and drainage systems. These requirements motivate the investigation of alternative computational approaches, including Physics-Informed Neural Networks.

### 2.1.3 Artificial Neural Networks

Artificial Neural Networks (ANNs) are computational models inspired by the structure of biological neural networks. They consist of interconnected processing units called neurons, which are organized into input, hidden, and output layers. Each neuron receives information, applies weights and biases, and passes the result through an activation function.

A general mathematical representation of a neural network is given by

$$\mathbf{y} = f_\theta(\mathbf{x}),$$

where $\mathbf{x}$ represents the input variables, $\mathbf{y}$ represents the predicted output, $f_\theta$ denotes the neural network function, and $\theta$ represents the trainable parameters of the network.

During training, the network adjusts its parameters to minimize a loss function that measures the difference between predicted and reference values. Optimization algorithms such as gradient descent and its variants are commonly used to update the network parameters.

ANNs can approximate complex nonlinear relationships between input and output variables. In flood prediction, they may be trained using rainfall, water level, terrain, or historical flood observations to estimate flood-related variables.

However, a purely data-driven ANN depends heavily on the quality and quantity of its training data. Its predictions may not automatically satisfy the physical laws governing water flow. This limitation provides the motivation for incorporating physical equations into the neural network training process.

### 2.1.4 Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) are neural networks that incorporate governing physical equations into their training process. Rather than relying exclusively on observed data, PINNs use mathematical equations to impose physical constraints on the predicted solutions.

Raissi, Perdikaris, and Karniadakis (2019) introduced a framework in which differential equations are incorporated into neural network loss functions. Through this approach, the network is trained to approximate solutions that are consistent with both available data and the governing physical laws.

A PINN generally consists of a neural network that approximates the solution of a physical system and a loss function that includes the residuals of the governing equations. The residuals measure how closely the predicted values satisfy the mathematical equations.

For flood modeling, the Shallow Water Equations can be incorporated into the PINN loss function. The network can then be trained to predict water depth and velocity while minimizing the residuals of the equations describing the conservation of mass and momentum.

One potential advantage of PINNs is their ability to combine observational data with physical constraints. This may be useful when available flood observations are limited. However, PINNs also have limitations, including training difficulties, sensitivity to loss weighting, and challenges in approximating discontinuous flow behavior. Therefore, their accuracy and physical consistency must be evaluated rather than assumed.

## 2.2 Theoretical Background

This section presents the mathematical and computational theories that serve as the foundation of the proposed SWE-based PINN. It discusses the governing equations of shallow water flow, the principles of conservation of mass and momentum, and the mathematical structure of the PINN.

### 2.2.1 Shallow Water Equations

The Shallow Water Equations (SWE) are a system of partial differential equations used to describe the movement of water with a relatively small depth compared with its horizontal dimensions. They are derived from the principles of conservation of mass and momentum and are commonly used in flood modeling.

The two-dimensional SWE represent water flow through the horizontal directions while accounting for changes in water depth, terrain elevation, rainfall, and friction. The equations are commonly derived from the Navier-Stokes equations under assumptions such as hydrostatic pressure distribution and depth-averaged horizontal velocity.

In the present study, the SWE will serve as the governing mathematical model for describing flood behavior in the selected study area of Pasig City. The equations will be incorporated into the Physics-Informed Neural Network to constrain its predictions.

### 2.2.2 Conservation of Mass

The principle of conservation of mass states that mass cannot be created or destroyed within a system, although it may enter or leave the system through external sources or boundaries.

For shallow water flow, conservation of mass is represented by the continuity equation. In two dimensions, the equation may be expressed as

$$\frac{\partial h}{\partial t} + \frac{\partial (hu)}{\partial x} + \frac{\partial (hv)}{\partial y} = R - D,$$

where $h$ is the water depth, $u$ and $v$ are the horizontal velocity components, $R$ is the rainfall input, and $D$ is a simplified drainage-removal rate.

The term $\frac{\partial h}{\partial t}$ represents the rate of change of water depth over time. The terms $\frac{\partial (hu)}{\partial x}$ and $\frac{\partial (hv)}{\partial y}$ represent the changes in water flux in the horizontal directions.

The rainfall term $R$ accounts for the addition of water to the surface. The drainage term $D$ represents the removal of water from the modeled surface domain. For consistency, $R$ and $D$ must use compatible units, such as meters per second.

In this study, drainage will be represented through a simplified removal term when sufficient data are available. This representation does not constitute a complete simulation of an underground drainage network. Detailed pipe flow, manhole interactions, and surface-to-sewer exchange may require a separate coupled drainage model.

### 2.2.3 Conservation of Momentum

The principle of conservation of momentum describes how the movement of water changes due to forces acting on the flow. These forces include gravity, pressure gradients, terrain slope, and friction.

The two-dimensional momentum equations for shallow water flow may be expressed in conservative form as

$$\frac{\partial (hu)}{\partial t} + \frac{\partial}{\partial x}\left(hu^2 + \frac{1}{2}gh^2\right) + \frac{\partial (huv)}{\partial y} = -gh\frac{\partial z}{\partial x} + S_{fx},$$

and

$$\frac{\partial (hv)}{\partial t} + \frac{\partial (huv)}{\partial x} + \frac{\partial}{\partial y}\left(hv^2 + \frac{1}{2}gh^2\right) = -gh\frac{\partial z}{\partial y} + S_{fy},$$

where $g$ is the gravitational acceleration, $z$ is the terrain elevation, and $S_{fx}$ and $S_{fy}$ are the friction source terms in the respective directions.

The left-hand side of each equation describes the temporal and spatial changes in momentum. The right-hand side represents the forces acting on the water.

The terrain slope terms describe the influence of ground elevation on water movement. The friction source terms represent resistance caused by the ground surface. Friction may be represented using an appropriate formulation, such as Manning's equation, depending on the model configuration and the availability of surface roughness data.

In the proposed study, these equations will provide the physical constraints for the prediction of horizontal water velocity and flood behavior.

### 2.2.4 Governing Equations for Flood Flow

The governing equations of the proposed model consist of the continuity equation and the two momentum equations. Together, they represent the conservation of mass and momentum in two-dimensional shallow water flow.

The system may be written as

$$\begin{cases} \dfrac{\partial h}{\partial t} + \dfrac{\partial (hu)}{\partial x} + \dfrac{\partial (hv)}{\partial y} = R - D, \\ \dfrac{\partial (hu)}{\partial t} + \dfrac{\partial}{\partial x}\left(hu^2 + \dfrac{1}{2}gh^2\right) + \dfrac{\partial (huv)}{\partial y} = -gh\dfrac{\partial z}{\partial x} + S_{fx}, \\ \dfrac{\partial (hv)}{\partial t} + \dfrac{\partial (huv)}{\partial x} + \dfrac{\partial}{\partial y}\left(hv^2 + \dfrac{1}{2}gh^2\right) = -gh\dfrac{\partial z}{\partial y} + S_{fy}. \end{cases}$$

The system describes the changes in water depth and horizontal velocity over the computational domain.

The terrain elevation function $z(x, y)$ provides information about the ground surface. Rainfall $R(x, y, t)$ represents the water input, while $D(x, y, t)$ may be used to represent a simplified drainage-removal rate. Initial and boundary conditions are also necessary to define the state of the water system at the beginning of the simulation and along the boundaries of the modeled area.

The governing equations will be used to construct the physics-based residuals of the proposed PINN.

### 2.2.5 Physics-Informed Neural Network Framework

The proposed PINN will approximate the solution of the governing equations using a neural network. The network will receive spatial, temporal, and environmental variables as inputs and generate estimates of water depth and horizontal water velocity.

The proposed input and output variables are represented as

$$\mathbf{X} = (x, y, t, R, z, D),$$

and

$$\mathbf{Y}_\theta = (h_\theta, u_\theta, v_\theta),$$

where $\theta$ represents the trainable parameters of the neural network.

The input variables describe the location, time, rainfall, terrain elevation, and simplified drainage representation. The output variables represent the predicted water depth and the horizontal velocity components.

The network may be expressed as

$$\mathbf{Y}_\theta = f_\theta(x, y, t, R, z, D).$$

The governing equations are incorporated into the training process through their residuals. These residuals are calculated using automatic differentiation, which allows the neural network to compute the required derivatives of its predicted outputs with respect to spatial and temporal coordinates.

The continuity residual may be defined as

$$r_c = \frac{\partial h_\theta}{\partial t} + \frac{\partial (h_\theta u_\theta)}{\partial x} + \frac{\partial (h_\theta v_\theta)}{\partial y} - R + D.$$

The momentum residuals may be defined as

$$r_x = \frac{\partial(h_\theta u_\theta)}{\partial t} + \frac{\partial}{\partial x}\left(h_\theta u_\theta^2 + \frac{1}{2}gh_\theta^2\right) + \frac{\partial(h_\theta u_\theta v_\theta)}{\partial y} + gh_\theta\frac{\partial z}{\partial x} - S_{fx},$$

and

$$r_y = \frac{\partial(h_\theta v_\theta)}{\partial t} + \frac{\partial(h_\theta u_\theta v_\theta)}{\partial x} + \frac{\partial}{\partial y}\left(h_\theta v_\theta^2 + \frac{1}{2}gh_\theta^2\right) + gh_\theta\frac{\partial z}{\partial y} - S_{fy}.$$

The residuals measure the extent to which the network's predictions satisfy the governing equations. Ideally, the residuals approach zero at the sampled points within the computational domain.

### 2.2.6 Loss Function of a PINN

The loss function is an essential component of the PINN because it determines how the network parameters are updated during training. The proposed model will use a composite loss function that combines data-based and physics-based constraints.

The total loss function may be expressed as

$$\mathcal{L}(\theta) = \mathcal{L}_{data} + \lambda_{PDE}\mathcal{L}_{PDE} + \lambda_{BC}\mathcal{L}_{BC} + \lambda_{IC}\mathcal{L}_{IC},$$

where $\mathcal{L}_{data}$ is the data loss, $\mathcal{L}_{PDE}$ is the governing-equation residual loss, $\mathcal{L}_{BC}$ is the boundary condition loss, $\mathcal{L}_{IC}$ is the initial condition loss, and the $\lambda$ terms are weighting parameters.

The data loss measures the difference between the predicted values and the available reference observations. It may be represented using the mean squared error:

$$\mathcal{L}_{data} = \frac{1}{N_d}\sum_{i=1}^{N_d}\left\|\mathbf{Y}_\theta^{(i)} - \mathbf{Y}_{ref}^{(i)}\right\|^2,$$

where $N_d$ is the number of reference data points and $\mathbf{Y}_{ref}$ represents the observed or reference solution.

The physics-based loss measures the residuals of the governing equations:

$$\mathcal{L}_{PDE} = \frac{1}{N_f} \sum_{i=1}^{N_f} \left[ (r_c^{(i)})^2 + (r_x^{(i)})^2 + (r_y^{(i)})^2 \right],$$

where $N_f$ is the number of collocation points sampled within the computational domain.

The boundary condition loss measures the difference between the network predictions and the prescribed boundary values:

$$\mathcal{L}_{BC} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left\| \mathbf{Y}_\theta^{(i)} - \mathbf{Y}_{BC}^{(i)} \right\|^2.$$

The initial condition loss measures the difference between the predicted and prescribed initial states:

$$\mathcal{L}_{IC} = \frac{1}{N_i} \sum_{i=1}^{N_i} \left\| \mathbf{Y}_\theta(x_i, y_i, 0) - \mathbf{Y}_{IC}^{(i)} \right\|^2.$$

The weighting parameters determine the relative importance of the individual loss components. Appropriate weighting is necessary because the terms may have different numerical scales. The loss function will guide the network toward approximating the reference solution while satisfying the physical constraints of water flow.

## 2.3 Review of Related Literature

This section discusses existing literature and theoretical discussions related to flood prediction, hydrodynamic modeling, the Shallow Water Equations, and the application of machine learning, hybrid models, and Physics-Informed Neural Networks.

### 2.3.1 Flood Prediction and Modeling

Flood prediction and modeling involve estimating the behavior of floodwater under specified environmental and hydrological conditions. These approaches may be used to estimate water depth, flow velocity, inundation extent, and the temporal development of flooding.

Traditional flood modeling commonly relies on hydrological and hydrodynamic methods. Hydrological models describe the transformation of rainfall into runoff, while hydrodynamic models represent the movement of water through rivers, channels, and surface areas. The selection of a modeling approach depends on the purpose of the study, the available data, and the level of detail required.

Guo et al. (2021) reviewed urban surface water flood modeling approaches and discussed the use of drainage network models, shallow-water-based models, hydro-geomorphic approaches, and other methods. Their review emphasized the importance of considering computational efficiency, model complexity, and data availability when selecting a flood modeling approach.

More recent literature has focused on the trade-off between model realism and speed. Perera et al. (2026) reported that traditional hydrodynamic models are based on physical laws and are therefore realistic and explainable, but can require considerable time and computing power. Machine learning models, in contrast, can be much faster, which is useful for real-time warnings, but depend heavily on data and can be difficult to interpret. The timing issue is practical: Besseling et al. (2024) noted that a hydrodynamic simulation of a large area can take 6 to 10 hours, whereas emergency decisions may require predictions within seconds or minutes.

The literature indicates that flood prediction requires a balance between computational efficiency and the ability to represent physical processes. This consideration is relevant to the present study, which investigates the use of a PINN as an alternative computational approach for solving the governing equations of flood flow.

### 2.3.2 Flood Modeling in Urban Areas

Urban flood modeling presents additional challenges because of the complex characteristics of developed environments. Buildings, roads, drainage networks, and variations in terrain elevation influence the direction and accumulation of water. Impermeable surfaces may also increase surface runoff during rainfall events.

Guo et al. (2021) explained that urban surface water flood models may use simplified or full forms of the Shallow Water Equations. The complexity of the model affects the required input data, computational cost, and the amount of flood information that can be represented.

Urban flood models may also consider the interaction between surface water and underground drainage systems. However, detailed representations of drainage networks require information about pipes, inlets, outlets, and other infrastructure. The availability of these data may affect the complexity and reliability of the model. In the study of Liu et al. (2026), which coupled a stormwater drainage model with a surface-flow model, elevation-related features and drainage features such as manholes were among the most influential variables explaining predicted flood depth. This supports the consideration of terrain and drainage in urban flood modeling, while also showing that such studies depend on high-quality spatial data.

For the present study, urban flood modeling will focus on the selected area of Pasig City. Terrain elevation and rainfall will be considered as important inputs. Drainage may be represented through a simplified removal term, subject to the availability of suitable data. The study will not attempt to fully simulate every component of the underground drainage infrastructure.

### 2.3.3 Shallow Water Equation Applications

The Shallow Water Equations are widely applied in flood modeling because they describe the conservation of mass and momentum in depth-averaged water flow. They can be used to estimate the movement and distribution of water over terrain.

SWE-based models may be implemented using different numerical methods, including finite difference, finite volume, and finite element approaches. These methods approximate the governing equations using discrete computational representations.

Guo et al. (2021) discussed the application of shallow-water-based models to urban surface flooding and noted their ability to provide spatially distributed information about floodwater depth and velocity. The review also highlighted that the computational cost and data requirements of such models may increase when detailed urban features are considered.

The mathematical form in which the SWE are written can also matter when they are solved with neural networks. Tian et al. (2025) compared the conservation form, the primitive variables form, and a variable-conservation form of the two-dimensional SWE in a PINN setting and found that the choice of formulation influenced optimization and approximation accuracy.

The SWE are relevant to the present study because they provide the physical basis for the proposed PINN. By incorporating the equations into the neural network loss function, the model will be trained to approximate flood behavior while considering the governing principles of water flow.

### 2.3.4 Physics-Informed Neural Networks in Hydrology

Physics-Informed Neural Networks have been investigated as a computational approach for solving differential equations and incorporating physical knowledge into machine learning models. Raissi et al. (2019) introduced a framework for forward problems, in which the solution of a known partial differential equation is approximated, and inverse problems, in which unknown parameters or terms of the equation are inferred from data. In their continuous-time formulation, a neural network approximates the solution, automatic differentiation computes the required derivatives, and the network is trained on a loss that combines data, initial condition, boundary condition, and equation-residual

terms. They also proposed a discrete-time formulation that incorporates implicit Runge-Kutta schemes.

In flood-related applications, PINNs can be used to represent processes governed by mathematical equations, such as water flow and rainfall-driven runoff. Perera et al. (2026) found that hybridization of flood models can occur at the input, the model structure, or the processing and output stages, and that PINNs, which modify the model structure by embedding physical laws, are especially promising because they may yield more physically realistic and generalizable predictions.

However, the use of PINNs also presents challenges. Raissi et al. (2019) showed that performance can depend on the network architecture, training strategy, placement of collocation points, and weighting of the loss components, and that the method can become computationally expensive for large domains. Tian et al. (2025) likewise reported issues of training instability, loss imbalance, and sensitivity to the equation formulation in shallow water problems. Therefore, the performance of a PINN depends not only on the physical equations but also on its architecture, training configuration, and evaluation process.

The present study applies this framework to the two-dimensional Shallow Water Equations. The governing equations will be incorporated into the PINN loss function to constrain the predicted water depth and horizontal velocity.

### 2.3.5 Machine Learning for Flood Prediction

Machine learning approaches have been used in flood prediction to identify relationships between rainfall, water levels, and flood-related variables. Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks are examples of approaches that can learn patterns from historical or simulated data.

LSTM networks are designed to process sequential information (Houdt, Mosquera, & Nápoles, 2020). Their structure allows them to retain information from previous time steps, making them suitable for tasks involving rainfall sequences, water-level records,

and time-dependent flood behavior. Besseling et al. (2024) applied an LSTM to predict flood inundation after a dike breach. The network took the discharge through the breach as input and predicted water depth over a grid of 34,629 cells, using 73 scenarios generated by a hydrodynamic model. On unseen scenarios, the model matched the hydrodynamic results closely and ran in a fraction of a second, compared with 6 to 10 hours for the hydrodynamic model. The authors, however, noted that the model depended on simulation data, was location-specific, required many simulations, did not enforce physical laws explicitly, and performed poorly immediately after the onset of flooding.

Data-driven approaches may provide efficient predictions when sufficient training data are available. However, their predictions depend on the patterns learned from the training data, and they do not automatically satisfy the governing equations of water flow unless physical constraints are explicitly incorporated (Liu et al., 2026; Perera et al., 2026).

PINNs differ from purely data-driven models because they include physical equations as part of the training objective. For example, Praveen et al. (2024) reported that a PINN for real-time flood prediction outperformed ANN and LSTM models, while Liu et al. (2026) reported that a SWE-based PINN outperformed DNN, CNN, and LSTM models. This distinction is important in the present study, where the model will be evaluated based on both approximation error and physical consistency.

A classical numerical method will be used as the primary comparison method for assessing the PINN's approximation of the Shallow Water Equations. If an LSTM model is included as an additional baseline, its prediction accuracy may be evaluated using the same reference dataset and error metrics. The LSTM comparison will be treated as a separate data-driven evaluation.

### 2.3.6 Flood Risk in Pasig City and Metro Manila

Metro Manila is a highly urbanized region where flood behavior may be influenced by rainfall, river systems, terrain, and urban infrastructure. Flood modeling in the region requires consideration of both the hydrological conditions and the characteristics of the

built environment.

The study provides regional context for the present research because Pasig City is located within the broader Metro Manila urban environment associated with the Pasig River system. However, the results of a basin-level study cannot automatically be applied to every location within Pasig City. Local terrain, rainfall conditions, and available flood observations must be considered.

The present study will focus on the development and evaluation of a SWE-based PINN within a selected study area in Pasig City. Its findings will be limited to the modeled area and the datasets used in the implementation.

### 2.3.7 Emerging Computational Approaches for Flood Prediction

Beyond classical numerical and machine learning methods, other computational paradigms have been proposed for climate and flood problems. Ho et al. (2024) reviewed the potential of quantum computing for climate resilience and sustainability, discussing quantum optimization, quantum machine learning, and quantum simulation, including techniques such as the Quantum Approximate Optimization Algorithm, quantum annealing, quantum support vector machines, and quantum convolutional neural networks. They identified flood and disaster prediction as possible applications.

The authors also noted that most of these applications remain at an early stage. Current quantum hardware is limited by noise, scalability, and the number of available qubits, so hybrid quantum-classical approaches are currently more practical. The review also did not develop or test a specific model on a climate dataset, and few studies compare quantum approaches with strong classical and physics-informed models.

This literature is outside the scope of the present study, which uses classical computation. It is included to show that research on computational approaches for flood prediction is still developing and that physics-informed and hybrid methods such as PINNs are among the practical options currently available.

## 2.4 Review of Related Studies

This section presents previous research studies directly related to the proposed study. The discussion focuses on the methods, governing equations, datasets, findings, and limitations of the selected studies.

### 2.4.1 SWE-Based Flood Prediction Studies

SWE-based flood prediction studies commonly use numerical methods to solve the governing equations of water flow. These methods include finite difference, finite volume, and finite element approaches. They are used to approximate water depth and velocity over a computational domain.

Guo et al. (2021) conducted a comprehensive review of urban surface water flood modeling approaches. Their study discussed the application of full and simplified Shallow Water Equations in urban flood simulation. It also identified computational efficiency, model complexity, and the availability of high-quality input data as important challenges in urban flood modeling.

The review provides a foundation for the present study because it establishes the relevance of SWE-based approaches in representing urban surface water flooding. However, the review primarily examines existing modeling approaches rather than developing a PINN specifically for Pasig City.

The present study builds on the use of the SWE as the governing mathematical model and investigates whether a neural network can approximate its solutions while maintaining physical consistency.

### 2.4.2 PINN-Based Flood Prediction Studies

Liu et al. (2026) developed a Shallow Water Equation-based PINN (SWEPINN) for urban flood depth prediction that integrates data-driven information with physical constraints. The study was conducted in Hongshan District, Wuhan, China. A coupled stormwater

drainage and surface-flow simulation covering 50 rainstorm scenarios was used to construct the dataset. Nine spatial features and fourteen rainfall features were selected using geographic detectors and backward selection, the model was trained to predict maximum flood depth, hyperparameters were tuned using Bayesian optimization, and SHAP analysis was used to interpret feature contributions. The SWEPINN outperformed DNN, CNN, and LSTM models, achieving an $R^2$ of 0.952, a mean absolute error of 0.068, and a mean squared error of 0.013. Elevation and manhole features were among the most influential.

The study is particularly relevant to the present research because it directly applies a SWE-based PINN to urban flood depth prediction. However, it was developed and evaluated for a single urban area, relied on simulated rather than observed flood data, and required high-quality, location-specific spatial data. Differences in terrain, rainfall patterns, drainage infrastructure, and available observations may affect how its configuration transfers to Pasig City. The study also compared the SWEPINN with other machine learning models rather than analyzing its convergence and error against a classical numerical solver of the SWE.

Praveen et al. (2024) developed a PINN for real-time flood prediction using rainfall-runoff data. A Kaggle dataset of 50,000 rows and 21 columns was cleaned, split, and normalized, and features were selected using principal component analysis and a multi-feature algorithm. The PINN was trained using both data and physical constraints and was compared with ANN and LSTM models. It achieved an accuracy of 93.49%, a root mean square error of 0.3140, and a mean absolute error of 0.2169, outperformed both baselines, and could predict flood conditions with a lead time of at least one hour.

The study shows the potential of physics-informed models for real-time flood prediction. However, because it relied on a publicly available dataset, its real-world performance requires further validation, and testing across locations and extreme flood events was limited.

The present study will investigate the application of a SWE-based PINN to a selected

area of Pasig City. It will focus on the convergence and approximation error of the model and its ability to satisfy the conservation laws of water flow. The PINN will also be compared with a classical numerical method using an appropriate reference solution.

### 2.4.3 PINN Applications in Hydrodynamics

Tian et al. (2025) investigated the use of PINNs for solving the two-dimensional Shallow Water Equations with terrain topography and rainfall source terms. The PINN takes the spatial and temporal coordinates as inputs and predicts water depth and the two horizontal velocity components, with the governing equations, initial conditions, and boundary conditions included in the loss function. The researchers compared three equation formulations (conservation, primitive variables, and variable-conservation forms), imposed non-negative depth and dry-wet conditions, and tested an additional energy or entropy condition. To improve training, they used a dimensional transformation and neuron-wise local adaptive activation functions.

The PINN was evaluated on eleven cases: static water over different terrains, rainfall over flat and complex terrain, pseudo-two-dimensional and circular dam-break problems, tidal flow over uneven terrain, and tidal flow combined with rainfall. Results were compared with exact solutions or with classical schemes, namely the HLL scheme and an entropy-stable scheme, using mean absolute error and root mean square error. For the static terrain cases, errors were approximately in the range of $10^{-4}$ to $10^{-5}$. The variable-conservation form generally performed better than the primitive variables form. In the pseudo-two-dimensional dam-break case, the PINN solution was closer to the exact solution than the first-order HLL solution used in that comparison, although the authors emphasized that this does not show that PINNs are generally superior to classical methods. In the rainfall cases, errors did not necessarily accumulate over time as they can in time-stepping schemes, but they were larger at higher rainfall intensities. The additional entropy condition did not consistently improve results, caused excessive smoothing in the dam-break case, and could lead to training instability.

This study is directly relevant to the present research because it investigates the same governing mathematical system and compares a PINN with classical numerical solutions. It provides a basis for incorporating terrain and rainfall into the proposed PINN framework.

However, the cases considered by Tian et al. (2025) were controlled benchmark and synthetic cases designed to evaluate the mathematical capability of PINNs to solve the SWE. The study does not establish the performance of a model implemented in a real-world location such as Pasig City, and a more systematic comparison of convergence, approximation error, efficiency, and stability with classical methods across a wider range of conditions remains open. The present research will extend the investigation to a selected urban study area using the available local data.

The foundational work of Raissi et al. (2019) applied the PINN framework to problems including the Navier-Stokes, nonlinear Schrödinger, Allen-Cahn, and Korteweg-de Vries equations. Their results showed that PINNs could solve these nonlinear equations with relatively limited labeled data and could identify unknown parameters in inverse problems. They did not, however, provide a complete theoretical explanation of why training converges to accurate solutions, nor guarantees on convergence, stability, or error bounds. This leaves room for studies, such as the present one, that evaluate convergence and approximation error for a specific application.

### 2.4.4 Data-Driven and Hybrid Flood Prediction Studies

Besseling et al. (2024) developed an LSTM neural network to predict flood inundation after a dike breach. Because historical flood data for the study area were insufficient, 73 scenarios were generated using a one-dimensional and two-dimensional hydrodynamic model and divided into 43 training, 15 validation, and 15 testing scenarios. Hyperparameters were selected using Bayesian optimization over 50 combinations, with a best configuration of 928 LSTM neurons, a dropout of 0.25, and a learning rate of 0.001. On the unseen test scenarios, the model achieved a mean absolute error of 0.045 m (about

3% of the mean depth of 1.49 m), a root mean square error of 0.13 m, and a critical success index of 94% for flood extent. It required about 0.1 to 0.3 seconds per prediction compared with 6 to 10 hours for the hydrodynamic model. Its limitations were its dependence on simulated data, its location-specific nature, its need for many simulations, the absence of explicit physical constraints, and weak performance immediately after the breach.

This study illustrates the efficiency of data-driven models and the limitation that motivates PINNs: without explicit physics, a model learns patterns from data, including any errors in the simulations used to train it.

Perera et al. (2026) conducted a systematic review of hybrid flood inundation models. They searched Scopus and Web of Science, screened the results using defined criteria, and reviewed the selected journal articles, conference papers, reviews, and a book chapter in full. They found that hybrid models aim to combine the realism of hydrodynamic models with the speed of machine learning, and that PINNs are especially promising. They also found no consistent way of comparing hybrid flood models because studies use different datasets, scenarios, and evaluation metrics. Hybrid models still face data dependency, limited generalization to new situations, complexity, retraining requirements, and black-box behavior. The authors proposed a standardized benchmarking framework in which hybrid, hydrodynamic, and standalone machine learning models are tested under the same data, conditions, computational resources, and evaluation measures.

This review supports the design of the present study, which compares the PINN with a classical numerical method using the same reference solution and error metrics.

### 2.4.5 Flood Prediction Studies in the Philippines

Flood prediction studies in the Philippines have investigated rainfall-runoff modeling, flood inundation simulation, and the application of hydrodynamic models to river basins and urban environments.

The study is relevant because it demonstrates the application of computational flood

modeling within a Philippine river basin. It also provides a regional reference for the present research, which focuses on flood modeling in Pasig City.

Nevertheless, the study used a different modeling framework from the proposed SWE-based PINN. It did not specifically investigate the convergence, approximation error, and physical residuals of a PINN implemented in Pasig City. This difference supports the need to evaluate the proposed approach within the selected study area.

## 2.5 Synthesis of Literature

The reviewed literature establishes that flood prediction requires the representation of physical processes that govern the movement and accumulation of water. Traditional hydrodynamic models use mathematical equations and numerical methods to simulate water flow. Among these, the Shallow Water Equations are widely used because they describe the conservation of mass and momentum in depth-averaged water flow.

The literature also shows that urban flood modeling presents challenges related to complex terrain, drainage systems, input data availability, and computational efficiency. These challenges are particularly relevant when modeling flood behavior in highly developed areas.

Artificial Neural Networks and other machine learning approaches provide an alternative means of estimating flood-related variables from data. However, purely data-driven models may not automatically satisfy the physical laws governing water flow. Physics-Informed Neural Networks address this limitation by incorporating governing equations into the training process.

Previous studies by Raissi et al. (2019), Liu et al. (2026), and Tian et al. (2025) establish the theoretical and computational basis for using PINNs to solve differential equations and model flood behavior. Meanwhile, the Marikina-Pasig River Basin study provides a regional reference for flood modeling in the Philippines.

Despite these contributions, the performance of a SWE-based PINN depends on the

study area, available data, governing equation formulation, training process, and evaluation method. Therefore, the present study will develop and evaluate a PINN for a selected area of Pasig City and compare its predictions with a classical numerical method.

## 2.6 Research Gap

Existing studies have demonstrated the use of the Shallow Water Equations in flood modeling and the potential of Physics-Informed Neural Networks in solving nonlinear partial differential equations. Liu et al. (2026) investigated a PINN framework for urban flood depth prediction, while Tian et al. (2025) examined the use of PINNs for solving the two-dimensional SWE with terrain topography and rainfall source terms.

However, the results of these studies were obtained using their respective study areas, datasets, and model configurations. Their findings cannot automatically establish the performance of a SWE-based PINN when implemented in Pasig City.

The present study addresses this gap by developing and evaluating a SWE-based PINN for flood modeling in a selected area of Pasig City. It will examine the model's convergence, approximation error, and physical consistency, particularly its ability to satisfy the conservation of mass and momentum.

The study will also compare the PINN with a classical numerical method using appropriate error metrics. This comparison will provide an assessment of how closely the PINN approximates the reference solution and whether its predictions satisfy the governing equations.

The research gap is therefore centered on the evaluation of the proposed SWE-based PINN within the selected Pasig City study area, particularly in terms of convergence, approximation error, and physical constraint satisfaction.

## 2.7 Relevance to the Present Study

The reviewed literature and studies provide the mathematical, computational, and scientific foundation for the present research.

The Shallow Water Equations provide the governing mathematical model for describing water depth and horizontal velocity. The principles of conservation of mass and momentum establish the physical constraints that the proposed PINN must satisfy.

The work of Raissi et al. (2019) provides the theoretical foundation for incorporating differential equations into neural network training. The study of Liu et al. (2026) provides a direct reference for applying PINNs to urban flood depth prediction. Meanwhile, Tian et al. (2025) provides a relevant mathematical and computational reference for solving the two-dimensional SWE with terrain topography and rainfall source terms.

The reviewed urban flood modeling literature also establishes the importance of considering terrain, rainfall, drainage representation, and data availability. These considerations will guide the selection and preparation of the input data for the Pasig City implementation.

The present study will use these foundations to develop a PINN that predicts water depth and horizontal velocity. Its performance will be evaluated through convergence, approximation error, and physical consistency, and compared with a classical numerical method. Through this evaluation, the study aims to examine the applicability of a physics-informed computational approach to flood modeling in the selected study area.

## 2.8 Chapter Summary

This chapter presented the fundamental concepts, mathematical theories, related literature, and previous studies relevant to the development of a Shallow Water Equation-based Physics-Informed Neural Network for flood modeling in Pasig City.

The discussion introduced flooding, hydrodynamic modeling, artificial neural networks, and Physics-Informed Neural Networks. It also presented the Shallow Water

Equations, including the principles of conservation of mass and momentum, and explained how the governing equations can be incorporated into the PINN loss function.

The reviewed literature and studies established the relevance of SWE-based flood modeling and the use of PINNs in solving differential equations. They also identified challenges associated with urban flood modeling, including data availability, computational requirements, and the complexity of representing physical processes.

The research gap identified in this chapter concerns the development and evaluation of a SWE-based PINN for a selected area of Pasig City, particularly in terms of convergence, approximation error, and physical constraint satisfaction. These concepts and findings provide the foundation for the research methodology, model development, and evaluation procedures presented in the succeeding chapter.

# Chapter 3

# Research Methodology

This chapter presents the research methodology that will be followed in developing and evaluating the Shallow Water Equation-based Physics-Informed Neural Network (SWE-based PINN) for flood prediction in a selected study area in the Philippines. It discusses the research design, study location, data sources, model parameters, mathematical formulation, neural network architecture, data preprocessing, training procedure, validation methods, and data analysis procedures.

## 3.1 Research Methodology

This study will employ a computational and mathematical modeling approach to develop a Physics-Informed Neural Network based on the Shallow Water Equations. The proposed model will incorporate physical laws governing water flow into the neural network training process to predict water depth and horizontal water velocity.

The developed model will be evaluated by examining its convergence, approximation error, and consistency with the governing physical equations. Its performance will also be compared with a classical numerical method for solving the Shallow Water Equations.

## 3.2 Research Design

This study will utilize a quantitative, computational, and model-development research design. It will involve the mathematical formulation of the Shallow Water Equations, implementation of a Physics-Informed Neural Network, and numerical experiments to assess the model's performance.

The study will use quantitative evaluation metrics to measure prediction accuracy, convergence behavior, and physical consistency. Comparisons with a classical numerical method will be conducted to determine the performance of the proposed model.

## 3.3 Research Locale and Period

The study will focus on selected flood-prone areas in Pasig City, Philippines, particularly areas associated with the Pasig River and its relevant waterways.

The research will cover the period from [Start Date] to [End Date]. The selected study area and simulation domain will be determined based on the availability of geographical, rainfall, and hydrological datasets required for the mathematical model.

## 3.4 Sources of Data

The study will utilize secondary data obtained from relevant government agencies, public databases, and other reliable sources. These datasets will provide the necessary information for developing, training, and evaluating the proposed model.

The potential data requirements include:

- **Rainfall Data:** Historical rainfall measurements from relevant meteorological monitoring stations.
- **Water-Level Data:** Historical water-level measurements from the Pasig River and other relevant waterways.
- **River Flow and Discharge Data:** Available measurements describing the movement and volume of water within the selected waterways.
- **Geographic and Elevation Data:** Digital Elevation Models (DEM) and other geographical information describing the terrain of the study area.
- **Historical Flood Observations:** Available records of flood occurrences, water depths, durations, and affected locations.

- **Initial and Boundary Conditions:** Data required to establish the initial state of the water system and the conditions along the boundaries of the simulation domain.

The final selection of datasets will depend on their availability, accessibility, completeness, and suitability for the proposed mathematical model.

## 3.5 Model Parameters, Variables, and Initial/Boundary Conditions

This section identifies the variables and parameters used in formulating and implementing the SWE-based PINN.

The model will consider the following variables:

- **Spatial Coordinates:** The horizontal coordinates representing the position within the study area.
- **Time:** The temporal variable representing the progression of the flood simulation.
- **Water Depth:** The depth of water at a given spatial location and time.
- **Horizontal Water Velocity:** The velocity components describing water movement in the horizontal directions.
- **Rainfall:** Precipitation data that may contribute to the source terms of the governing equations.

The model parameters will include relevant physical quantities and numerical settings, such as gravitational acceleration, terrain elevation, friction coefficients, and other parameters required by the governing equations.

Initial and boundary conditions will be established using available hydrological and geographical information to define the behavior of the water system within the simulation domain.

## 3.6 Mathematical Model Formulation

This section presents the mathematical foundation of the proposed model using the Shallow Water Equations.

### 3.6.1 Governing Shallow Water Equations

Present the system of partial differential equations describing the movement of water across the selected study area.

### 3.6.2 Conservation of Mass

Discuss the continuity equation governing the conservation of water mass within the simulation domain.

### 3.6.3 Conservation of Momentum

Present the momentum equations describing changes in water velocity due to gravitational forces, terrain slope, and friction.

### 3.6.4 Initial Conditions

Define the initial water depth and horizontal velocity conditions at the beginning of the simulation.

### 3.6.5 Boundary Conditions

Specify the conditions imposed along the boundaries of the computational domain, including relevant inflow, outflow, or other boundary constraints.

## 3.7 PINN Model Architecture

This section describes the architecture and mathematical structure of the Physics-Informed Neural Network used to approximate the solutions of the Shallow Water Equations.

### 3.7.1 Neural Network Structure

Describe the neural network architecture, including its input layer, hidden layers, activation functions, and output layer.

### 3.7.2 Input and Output Variables

Identify the spatial coordinates and time as the primary network inputs. The outputs will represent the predicted water depth and horizontal water velocity components.

### 3.7.3 Physics-Based Constraints

Explain how the Shallow Water Equations are incorporated into the network using automatic differentiation to calculate the partial derivatives required for the governing equations.

### 3.7.4 Loss Function Formulation

Present the composite loss function, including the data loss, initial condition loss, boundary condition loss, and physics-based residual loss.

Explain how the different loss components are combined and minimized during training to obtain physically consistent predictions.

## 3.8 Data Preprocessing

This section discusses the procedures applied to the collected datasets before they are used in model development and training.

The preprocessing procedures may include:

- Data cleaning and removal of invalid observations.
- Normalization or scaling of numerical variables.
- Identification and handling of missing data.
- Spatial and temporal alignment of datasets.
- Preparation of training, validation, and testing datasets.

These procedures will ensure that the collected data are suitable for numerical computation and neural network training.

## 3.9 Model Training Procedure

This section explains the process of training the SWE-based PINN.

The training procedure will involve defining the network architecture, initializing its parameters, calculating the composite loss function, and updating the network weights through an optimization algorithm.

The following aspects will be discussed:

- Training and validation data preparation.
- Selection of optimization algorithms.
- Configuration of learning rates and training iterations.
- Selection of collocation points for evaluating the governing equations.
- Monitoring and minimization of the composite loss function.
- Evaluation of convergence during training.

## 3.10 Model Validation and Evaluation

The developed SWE-based PINN will be evaluated to determine its accuracy, convergence behavior, and physical consistency.

The evaluation will consider the following metrics:

- **Root Mean Square Error (RMSE):** Measures the average magnitude of prediction errors while giving greater weight to larger errors.
- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and reference values.
- **Coefficient of Determination ($R^2$):** Measures how well the model explains the variation in the reference data, where appropriate.
- **Physics-Based Residual Error:** Measures the extent to which the predicted solutions satisfy the governing Shallow Water Equations.
- **Convergence Analysis:** Examines the behavior of the model's loss function and approximation error throughout the training process.

The PINN predictions will be compared with available measured observations or numerical solutions obtained using a classical method for solving the Shallow Water Equations.

## 3.11 Data Analysis Procedure

The results generated by the proposed model will be analyzed using quantitative and numerical methods.

The analysis will focus on the following:

1. Examining the convergence behavior of the PINN during training.
2. Measuring the approximation error between the predicted and reference solutions.
3. Evaluating the physical consistency of the predicted water depth and velocity using the governing equations.
4. Comparing the performance of the PINN with a classical numerical method.
5. Interpreting the results to determine the effectiveness and limitations of the proposed model for flood prediction.

The findings will be presented using appropriate tables, graphs, and other visual representations to support the interpretation of the model's performance.

# Chapter 4

# How to Cite References

First edit the file `bibliog.bib` by adding an entry for each reference used in your thesis. Follow the format in the sample file.

```
@BOOK{bsw,
  author={H. T. Banks and R. C. Smith and Y. Wang},
  title={Smart Material Structures: Modeling Estimation
         and Control},
  year=1996,
  publisher={Masson/John Wiley},
  address={Paris/Chichester}
}

@ARTICLE{BanksIto1,
  author={H. T. Banks and K. Ito},
  year=1997,
  title={Approximation in {LQR} Problems for Infinite Dimensional
    Systems with Unbounded Input Operators},
  journal={Journal of Mathematical Systems, Estimation, and Control},
  volume=7,
  pages={1-34},
  publisher={Birkh\"{a}user Boston}
}

@PHDTHESIS{Ric,
```

```
  author = {R. C. H. del Rosario },
  year = 1998,
  title = {Computational Methods for Feedback Control in
        Structural Systems},
  school = {N. C. State University}
}

@techreport{RalphRic3a,
  author = {R. C. H. del Rosario and R. C. Smith},
  year = 1997,
  title = {{LQR} Control of Thin Shell Dynamics: Formulation and
           Numerical Implementation},
  institution = {Institute for Computer Applications in Science
    and Engineering},
  number={ICASE Report 97-59},
  address={NASA Langley Research Center, Hampton},
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%   %MISC   %%%%%%%%%%%%%%%%%%%%%%%%%%%%

@BOOKLET{Le,
  author={A. W. Leissa},
  title={Vibrations of Shells},
  year=1973,
  publisher={NASA SP-288},
  howpublished={Reprinted by the Acoustical Society of America
   through the American Institute of Physics}
}
```

```
@manual{d31ref,
  organization = {{Institute} of Electrical and
   Electronics Engineers},
  year = 1978,
  title = {IEEE Standard on Piezoelectricity},
  address={New York}
}

@manual{pside,
  title = {Specification of PSIDE},
  author={Jacques J.B. de Swart and Walter M. Lioen and
   Wolter A. van der Veen},
  year=1998,
  note={Avalable at http://www.cwi.nl/cwi/projects/PSIDE/}
}
```

Click from the BibTex icon (if you are using WinEdit Program) or use the BibTex command if you are in command line mode. It is suggested that you do this twice. Then compile the main tex file using the LaTeX command.

Use the `cite` command to cite a reference. For example, `\cite{bsw}` cites the book labeled "bsw". This will give us (Banks, Smith, & Wang, 1996).

# Appendix A

# ThesisMain.tex File

The ThesisMain.tex file contains the basic information about your thesis while the chapter files are just called in the code. This is the only file you need to compile. Do not run the chapter files.

## A.1 Preamble

```
\documentclass[12pt]{report}
\usepackage{feumathstyle}
\usepackage{amssymb, graphicx, amsmath}
\usepackage{hyperref}
\usepackage{apacite}
\bibliographystyle{apacite}

\usepackage{lscape}

\newtheorem{definition}{Definition}[chapter]
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}
\newtheorem{proposition}[theorem]{Proposition}
```

you may add additional packages but do not remove the prelisted packages especially the feumathstyle package. You may add other new formats here too

## A.2 Basic Information

```
 \title{Thesis Title} %do not capitalize all


\authorfirst{Juan L.}
\authorlast{Dela Cruz}
\authorext{Jr.}
\authorexttrue %make this a comment if you do not have an extension name e.g. Jr.,

\secondauthorfirst{Juan L.}
\secondauthorlast{Dela Cruz}
\secondauthorext{Jr.}
\secondauthorexttrue %make this a comment if you do not have an extension name e.g

\thirdauthorfirst{Juan L.}
\thirdauthorlast{Dela Cruz}
\thirdauthorext{Jr.}
\thirdauthorexttrue %make this a comment if you do not have an extension name e.g.

\degree{Bachelor of Science in Applied Mathematics } %enter the specific degree
\thesistype{thesis}
\thesisdocument{undergraduate thesis}

\specialization{with Information Technology} \specializetrue

\defensedate{May 12, 2022} %indicate the month, day and year the research was defe
\submitmonth{May} %indicate the month the manuscript was submitted
\submityear{2022} %indicate the year the manuscript was submitted
```

```
\adviser{Adviser, Ph.D.}%main adviser
\panel{Panel, Ph.D.}
\secondpanel{Panel, Ph.D.}
\thirdpanel{Panel, Ph.D.}


\thesisinstructor{May Anne C. Tirado, Ph.D.}


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% highlight \bothabsenttrue if the student has only one adviser and one reader
%% otherwise comment \bothabsenttrue and highlight \coadviser or \coreader which
%% ever is applicable. In this case, the package assumes either you have 1 Adviser
%% and 2 Readers, or 2 Advisers and 1 Reader.

\bothabsenttrue

%\coadviser{co-Adviser, Ph.D.}
%\coadvisetrue
% Make the \coadvisetrue as a comment if there is no adviser
%%%%%%%%%%%%%%%%%%%%%%%%%%%


\deptchair{Genesis John G. Borja}
\dean{Dean Name}
```

input the names and information about your thesis, authors, adviser, panel, dean.

## A.3 Abstract and Acknowledgements

```
\abs{Type your abstract here.  Your abstract should have a minimum
 of 150 words and a maximum of 300 words.}

\acknowledge{Type your acknowledgements here. It is customary to acknowledge spec
 Acknowledgments should be couched in terms consistent with the
 scholarly nature of the work. Your name and the date should not appear on this p
```

type your abstract and acknowledgements here

## A.4 Chapters and Appendix

Type your chapters and appendix in separate tex files. This is so it will be easier to edit a shorter code than the full manuscript.

The chapter files starts with

```
\chapter{How to Wrtie Text and Type Equations}
\label{chap:typing}
```

The label is optional. This part is when you want to cite this chapter in other chapters. This file just ends with the last line of your chapter content.

## A.5 Document Outline

```
\beforepreface
\abstractpage

\acknowledgetrue
\acknowledgepage
```

```
\tableofcontents

\newpage


\listoftables
\listoffigures

\afterpreface

\include{1typing}
\include{2compiling}
\include{3figures}
\include{4citing}
%%%%% edit the chapter files. no more preamble. start directly on \chapter{}. add r

\appendix
\input{appendixA}
\input{appendixB}
%%%%% edit appendix files. no more preamble. start directly on \chapter. you may re

\nocite*
\bibliography{BIBLIOG.bib}
%to edit your bibliography, open the BIBLIOG.BIB file.
```

The first set of commands will create beforepreface using the information you inputed at the start of ThesisMain.tex. Include the chapters in order and input the appendices in order too. The chapter file names are used in the arguments. It is best to not use

space in the chapter file names.

## A.6 Bibliography

Input your references in the Bibliography.bib file

```
@article{Datta-Figueira-2013,
 author = {Dilip Datta and Jos\'e Rui Figueira},
 title = {{A real-integer-discrete-coded differential evolution}},
 journal = {Applied Soft Computing},
 volume = {13},
 number = {9},
 pages = {3884--3893},
 year = {2013}
}
@book{Deb-2001,
 author = {Kalyanmoy Deb},
 title = {{Multi-Objective Optimization using Evolutionary Algorithms}},
 publisher = {John Wiley \& Sons Ltd.},
 address = {Chichester, England},
 year = {2001}
}
```

## A.7 Folder Contents

All your thesis files must be in one folder before you compile/run ThesisMain.tex

1. ThesisMain.tex
2. feumathstyle.sty

3. FEUlogo.png
4. IASlogo.png
5. BIBLIOG.bib
6. chapter tex files
7. appendix tex files

See the sample compiled ThesisMain.pdf to see the intended output.

# Appendix B

# feumathstyle.sty file

DO NOT EDIT feumathstyle.sty unless necessary. If you want to edit beforepreface contents that are not on the main tex file, and other formats, do it on feumathstyle.sty file. Be careful in editing this file. You may create a copy first and edit the copy.

feumathstyle.sty contains the text format and layout of the beforepreface pages, chapter pages, appendices, table of contents and other lists.

the definition of some commands are also on this file.

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